

# GES Phase 09

## Enterprise Human Resources, Workforce & Talent Management

Version 1.0.0 | Review-ready | Proposed Lock Tag: GES-P09-v1.0.0

### Executive Summary

Phase 09 defines the construction-specific workforce operating system for GrounUp, connecting workforce demand, recruiting, onboarding, employee records, crews, time, labor compliance, payroll integration, training, safety, performance, career development, self-service, analytics and governed AI workforce intelligence.

### Controlled Inventory

| Artifact           | Count |
|--------------------|-------|
| Requirements       | 210   |
| Canonical entities | 94    |
| Business rules     | 20    |
| Validation rules   | 20    |
| Workflows          | 20    |
| Engines            | 24    |
| APIs               | 28    |
| UI surfaces        | 15    |
| Reports            | 15    |
| KPIs               | 25    |
| Test cases         | 210   |

### Functional Domains

#### Workforce Planning

Forecast labor demand, capacity, headcount, hiring and succession needs.

#### Recruiting and ATS

Manage requisitions, candidates, interviews, screening, offers and hiring.

#### Onboarding

Coordinate prehire compliance, orientation, documents, equipment and first assignments.

#### Employee Master

Maintain governed worker, employment, contact, classification and organizational records.

#### Skills and Competencies

Track proficiency, trade skills, machine qualifications and competency evidence.

## **Crew Management**

Define crew templates, assignments, leadership, shifts and project deployment.

## **Time and Attendance**

Capture geofenced, offline-capable time with approvals and job-cost allocation.

## **Workforce Scheduling**

Schedule shifts, rotations, availability, leave, call-outs and standby coverage.

## **Compensation and Payroll Integration**

Apply pay rules, premiums, deductions and payroll export controls.

## **Union and Labor Agreements**

Govern locals, dispatch, classifications, wage tables, fringes and agreement rules.

## **Prevailing Wage and Certified Payroll**

Determine wage decisions and produce auditable certified payroll data.

## **Training and LMS**

Assign, deliver and record training, assessments and continuing education.

## **Licenses and Certifications**

Track requirements, expirations, renewals and role/project eligibility.

## **Safety and Health Integration**

Connect incidents, medical restrictions, return-to-work and fitness-for-duty controls.

## **Leave and Absence**

Manage leave requests, balances, approvals, restrictions and coverage impacts.

## **Performance Management**

Manage goals, reviews, coaching, corrective actions and development plans.

## **Career and Succession**

Define career paths, mentorship, apprenticeship and succession readiness.

## **Employee and Manager Self-Service**

Provide secure role-based employee and management workflows.

## **Workforce Digital Twin**

Maintain current capability, readiness, cost, availability and risk representations.

## **AI Workforce Intelligence**

Forecast, recommend and explain governed workforce decisions with human approval.

# Workforce Digital Twin and AI Governance

The Workforce Digital Twin is a versioned representation of worker capability, readiness, cost, availability, assignments, project experience, credentials, safety restrictions, development and risk. AI may forecast and recommend but may not independently make consequential employment decisions. Recommendations require evidence, confidence, explanation, versioned model/prompt references, bias evaluation and human review.

## Business Rules

- GES-P09-BR-001** - Worker assignment requires active employment and all mandatory credentials.
- GES-P09-BR-002** - Crew assignments shall not overlap unless the assignment type explicitly permits overlap.
- GES-P09-BR-003** - Union classification and agreement rules shall be resolved using work location and effective date.
- GES-P09-BR-004** - Prevailing wage decisions shall retain the determination source, version and calculation evidence.
- GES-P09-BR-005** - Certified payroll output shall reconcile approved time, classifications, rates, fringes and deductions.
- GES-P09-BR-006** - No AI recommendation may directly hire, terminate, discipline or change compensation without authorized human approval.
- GES-P09-BR-007** - Geolocation shall be collected only under approved tenant policy and disclosed worker consent rules.
- GES-P09-BR-008** - Sensitive HR data shall be restricted by least privilege and field-level security.
- GES-P09-BR-009** - Time entries shall retain original, edited and approved values with actor and timestamp.
- GES-P09-BR-010** - Credential expiration shall automatically affect eligibility according to configurable grace rules.
- GES-P09-BR-011** - Fatigue risk shall be advisory unless a configured safety threshold requires escalation.
- GES-P09-BR-012** - Performance decisions shall distinguish objective project evidence from manager commentary.
- GES-P09-BR-013** - Candidate records shall follow configurable retention and disposition requirements.
- GES-P09-BR-014** - Worker digital twin snapshots shall be versioned and reproducible from source evidence.
- GES-P09-BR-015** - Payroll exports shall be immutable after transmission; corrections require adjustment batches.
- GES-P09-BR-016** - Leave approvals shall evaluate available balance, staffing impact and applicable policy.
- GES-P09-BR-017** - Equipment qualification shall be model/asset-class specific where required by tenant policy.
- GES-P09-BR-018** - Union dispatch requests shall preserve referral sequence, response and acceptance evidence.
- GES-P09-BR-019** - Cross-tenant workforce records shall never be visible without explicit authorized shared-service configuration.
- GES-P09-BR-020** - Every workforce state transition shall create an audit event.

## Validation Rules

- GES-P09-VR-001** - Worker identifier is required and unique within a tenant.
- GES-P09-VR-002** - Employment start date must precede end date.
- GES-P09-VR-003** - Effective-dated records may not contain ambiguous overlapping periods.
- GES-P09-VR-004** - Time entry duration must be positive and within configured maximum shift limits.
- GES-P09-VR-005** - Timecard approval requires all entries to have project and cost allocation where required.
- GES-P09-VR-006** - Credential issue date may not be after expiration date.
- GES-P09-VR-007** - Crew assignment requires a valid project, shift and responsible supervisor.
- GES-P09-VR-008** - Union local must match an active labor agreement for union-rated work.
- GES-P09-VR-009** - Prevailing wage classification must map to an approved determination classification.
- GES-P09-VR-010** - Certified payroll week ending date must align with tenant payroll calendar.
- GES-P09-VR-011** - Candidate consent is required before regulated screening.
- GES-P09-VR-012** - Offer compensation must pass configured approval thresholds.

**GES-P09-VR-013** - Leave request end must not precede start.

**GES-P09-VR-014** - Performance review period must be closed before finalization.

**GES-P09-VR-015** - Medical restriction details must use restricted security classification.

**GES-P09-VR-016** - Payroll export totals must reconcile to approved source records.

**GES-P09-VR-017** - Offline time entries must include device timestamp and synchronization provenance.

**GES-P09-VR-018** - AI recommendations must include confidence, evidence and model/prompt version.

**GES-P09-VR-019** - Termination date requires final employment status and separation reason.

**GES-P09-VR-020** - Required onboarding tasks must be complete before activation unless waived by authorized role.

## Workflow Catalog

**GES-P09-WF-001 - Requisition to Hire:** Configured multi-step workflow for requisition to hire.

**GES-P09-WF-002 - Candidate Screening and Interview:** Configured multi-step workflow for candidate screening and interview.

**GES-P09-WF-003 - Offer and Prehire Compliance:** Configured multi-step workflow for offer and prehire compliance.

**GES-P09-WF-004 - New Hire Onboarding:** Configured multi-step workflow for new hire onboarding.

**GES-P09-WF-005 - Employee Status Change:** Configured multi-step workflow for employee status change.

**GES-P09-WF-006 - Crew Request and Assignment:** Configured multi-step workflow for crew request and assignment.

**GES-P09-WF-007 - Daily Time Capture and Approval:** Configured multi-step workflow for daily time capture and approval.

**GES-P09-WF-008 - Payroll Preparation and Export:** Configured multi-step workflow for payroll preparation and export.

**GES-P09-WF-009 - Union Dispatch and Referral:** Configured multi-step workflow for union dispatch and referral.

**GES-P09-WF-010 - Prevailing Wage Determination:** Configured multi-step workflow for prevailing wage determination.

**GES-P09-WF-011 - Certified Payroll Generation:** Configured multi-step workflow for certified payroll generation.

**GES-P09-WF-012 - Training Assignment and Completion:** Configured multi-step workflow for training assignment and completion.

**GES-P09-WF-013 - Credential Renewal:** Configured multi-step workflow for credential renewal.

**GES-P09-WF-014 - Leave Request and Coverage:** Configured multi-step workflow for leave request and coverage.

**GES-P09-WF-015 - Performance Review Cycle:** Configured multi-step workflow for performance review cycle.

**GES-P09-WF-016 - Corrective Action and Appeal:** Configured multi-step workflow for corrective action and appeal.

**GES-P09-WF-017 - Return to Work:** Configured multi-step workflow for return to work.

**GES-P09-WF-018 - Career Development and Mentorship:** Configured multi-step workflow for career development and mentorship.

**GES-P09-WF-019 - Workforce Planning and Hiring Forecast:** Configured multi-step workflow for workforce planning and hiring forecast.

**GES-P09-WF-020 - Offboarding and Access Revocation:** Configured multi-step workflow for offboarding and access revocation.

## Engine Catalog

**GES-P09-ENG-001 - Workforce Demand Forecasting Engine:** Processes governed workforce inputs and produces explainable outputs for workforce demand forecasting engine.

**GES-P09-ENG-002 - Capacity Planning Engine:** Processes governed workforce inputs and produces explainable outputs for capacity planning engine.

**GES-P09-ENG-003 - Recruiting Match Engine:** Processes governed workforce inputs and produces explainable outputs for recruiting match engine.

**GES-P09-ENG-004 - Candidate Scoring Engine:** Processes governed workforce inputs and produces explainable outputs for candidate scoring engine.

**GES-P09-ENG-005 - Onboarding Orchestration Engine:** Processes governed workforce inputs and produces explainable outputs for onboarding orchestration engine.

**GES-P09-ENG-006 - Worker Eligibility Engine:** Processes governed workforce inputs and produces explainable outputs for worker eligibility engine.

**GES-P09-ENG-007 - Crew Optimization Engine:** Processes governed workforce inputs and produces explainable outputs for crew optimization engine.

**GES-P09-ENG-008 - Workforce Scheduling Engine:** Processes governed workforce inputs and produces explainable outputs for workforce scheduling engine.

**GES-P09-ENG-009 - Time and Attendance Rules Engine:** Processes governed workforce inputs and produces explainable outputs for time and attendance rules engine.

**GES-P09-ENG-010 - Geofence Validation Engine:** Processes governed workforce inputs and produces explainable outputs for geofence validation engine.

**GES-P09-ENG-011 - Labor Rate Resolution Engine:** Processes governed workforce inputs and produces explainable outputs for labor rate resolution engine.

**GES-P09-ENG-012 - Union Agreement Engine:** Processes governed workforce inputs and produces explainable outputs for union agreement engine.

**GES-P09-ENG-013 - Prevailing Wage Engine:** Processes governed workforce inputs and produces explainable outputs for prevailing wage engine.

**GES-P09-ENG-014 - Certified Payroll Engine:** Processes governed workforce inputs and produces explainable outputs for certified payroll engine.

**GES-P09-ENG-015 - Payroll Reconciliation Engine:** Processes governed workforce inputs and produces explainable outputs for payroll reconciliation engine.

**GES-P09-ENG-016 - Credential Compliance Engine:** Processes governed workforce inputs and produces explainable outputs for credential compliance engine.

**GES-P09-ENG-017 - Training Recommendation Engine:** Processes governed workforce inputs and produces explainable outputs for training recommendation engine.

**GES-P09-ENG-018 - Fatigue Risk Engine:** Processes governed workforce inputs and produces explainable outputs for fatigue risk engine.

**GES-P09-ENG-019 - Retention Risk Engine:** Processes governed workforce inputs and produces explainable outputs for retention risk engine.

**GES-P09-ENG-020 - Performance Insight Engine:** Processes governed workforce inputs and produces explainable outputs for performance insight engine.

**GES-P09-ENG-021 - Career Path Engine:** Processes governed workforce inputs and produces explainable outputs for career path engine.

**GES-P09-ENG-022 - Succession Planning Engine:** Processes governed workforce inputs and produces explainable outputs for succession planning engine.

**GES-P09-ENG-023 - Workforce Digital Twin Engine:** Processes governed workforce inputs and produces explainable outputs for workforce digital twin engine.

**GES-P09-ENG-024 - Workforce Notification Engine:** Processes governed workforce inputs and produces explainable outputs for workforce notification engine.

## Approval and Lock

Status is Review-ready. Product, HR, Operations, Payroll, Labor Compliance, Safety, Architecture, Security/Privacy, Data Governance and UAT approvals are required before lock as GES-P09-v1.0.0.